

College of Engineering

Coursework Submission Sheet

Coursework Title: Assignment 1	Student number: 2502923
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Module title: Simulation Based Product Design...	Degree course: MSc in Mechanical Engineering
Submission deadline: 4pm on 28 th February....	
Lecturer: Dr Alison Williams.....	

By submitting this coursework, I certify that this is all my own work.

Submission date4pm on 28th February

Student signature if this is a non-electronic submission.....

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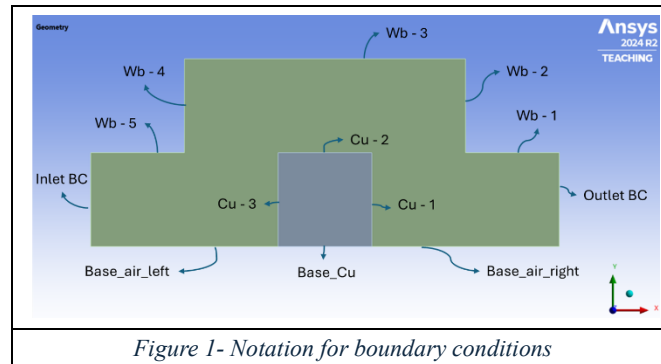
Mesh Sensitivity Study using ANSYS Fluent

1 Introduction

This report presents a mesh sensitivity study for a heat conduction case using ANSYS Fluent, with the objective of determining the maximum temperature in the system and evaluating the impact of mesh refinement on the outcomes.

The study involves generating five different meshes using the **sweep method**, with each mesh containing fewer than **100,000 elements** to maintain computational efficiency. The influence of mesh density on the simulation results is analysed to determine whether a mesh-independent solution can be achieved within the specified parameters.

2 Notations and abbreviations for Boundary conditions



Abbreviations*:

Wb – Wall boundary

Cu - Copper

*These abbreviations will be used to explain the edge sizing and biasing applied for different iterations.

3 Governing Equation for Heat Conduction

The governing equation for heat equation is:

$$\frac{\partial(\rho C_p T)}{\partial t} + \nabla \cdot (\rho \mathbf{v} C_p T) = \nabla \cdot (k \nabla T) + S_i$$

Where,

- ρ = density (kg/m³)
- C_p = specific heat capacity (J/kg·K)
- T = temperature (K)
- k = thermal conductivity (W/m·K)
- $\mathbf{v}$ = velocity vector (m/s)
- S_i = source term (W/m³), which accounts for the 2.5 W heat source in the copper block.

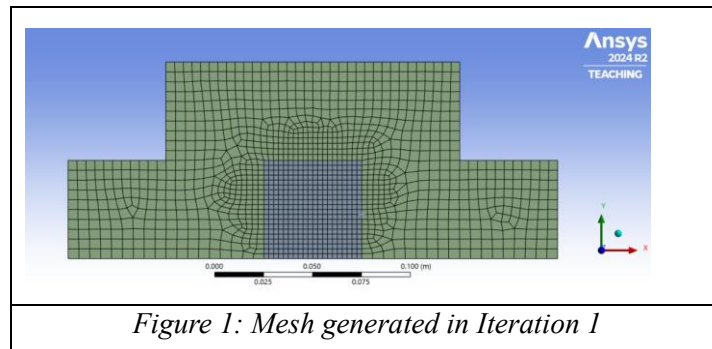
Since this is a pure conduction case, the velocity term $\mathbf{v}$ is zero, simplifying the equation to:

$$\frac{\partial(\rho C_p T)}{\partial t} = \nabla \cdot (k \nabla T) + S_i$$

For steady-state conditions:

$$\nabla \cdot (k \nabla T) + S_i = 0$$

3.1 Mesh 1



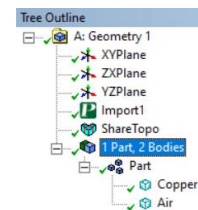
This initial structured mesh was selected as a reference to establish a baseline for the mesh sensitivity study. Further refinements were subsequently applied in later iterations to enhance solution accuracy and ensure mesh convergence.

3.1.1 Details of Mesh 1

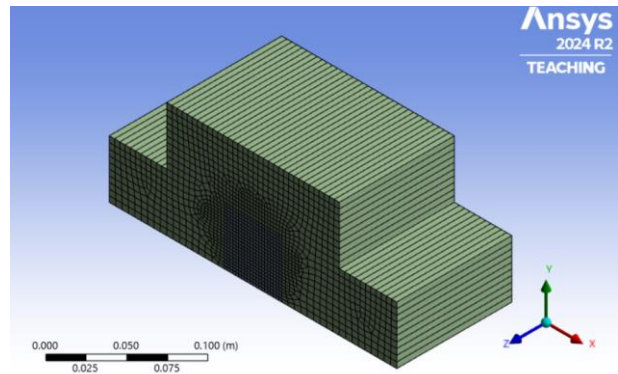
Mesh size (global)		0.005m
Mesh size (face sizing) copper block		0.0025m
Elements		1381
Skew	Min.	1.3057e-010
	Max.	0.56994
	Avg.	6.1395e-002
Orthogonal Quality:	Min.	0.69269
	Max.	1.
	Avg.	0.98932

3.1.2 Configurations set for Mesh 1

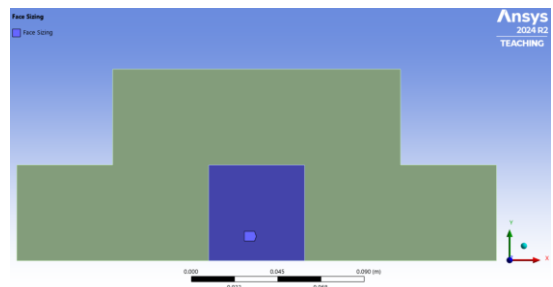
Model built as a part to facilitate topology sharing to avoid mesh discontinuity.



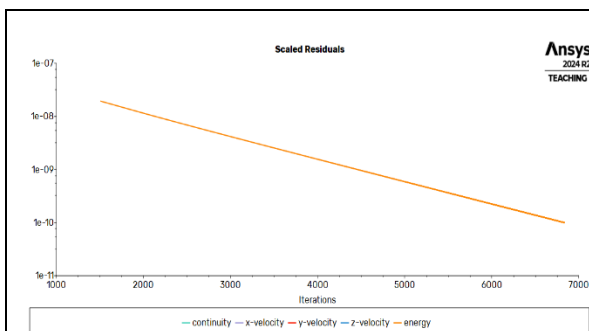
Sweep method used for both Copper and fluid interface.



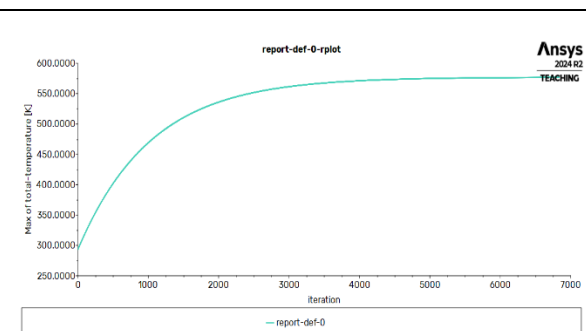
Face sizing of 0.0025m is given for Cu block



3.1.3 Results for Mesh 1



Residual plot for Mesh 1



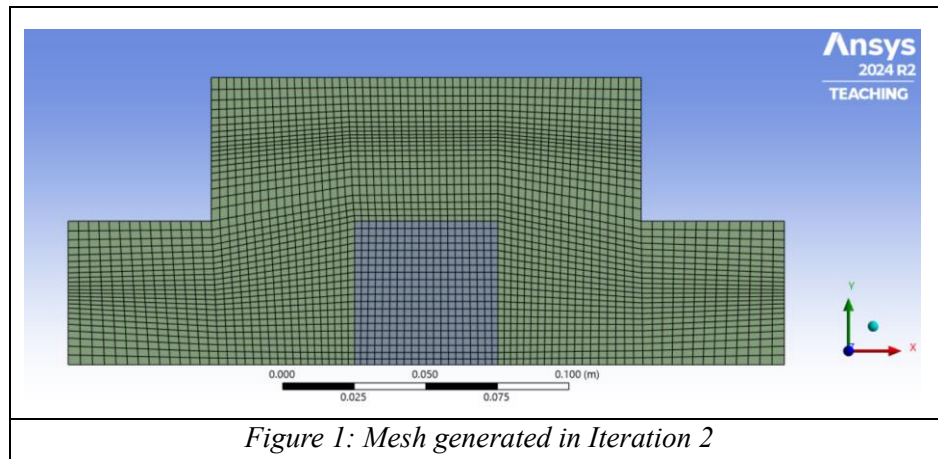
Temperature plot for Mesh 1

- Max. Temp: 5.7639e+02
- Iterations: 6135
- Scaled residual: 1e-10

Steps followed to ensure convergence:

- Larger element size for a smaller number of iterations and computational time.
- Residual set to 1e-10 to get a smoother graph and to check the convergence in the graph.

3.2 Mesh 2



In this iteration, the coarse mesh of iteration 1 was refined to give Quad/Hex elements by applying suitable edge sizing and biasing.

3.2.1 Details of Mesh 2

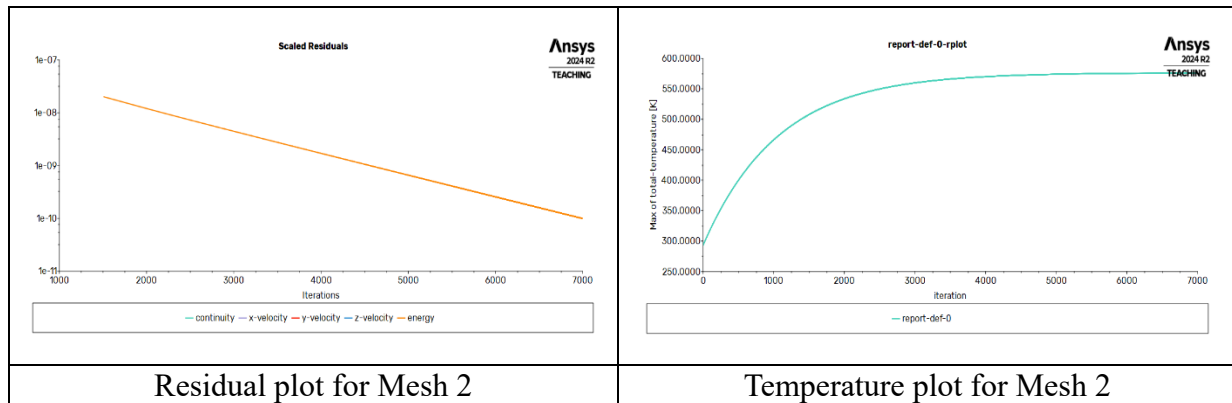
Mesh size (global)		0.005m
Mesh size (face sizing) copper block		0.0025m
Elements		3099
Skew	Min.	9.3154e-004
	Max.	0.13205
	Avg.	3.7849e-002
Orthogonal Quality:	Min.	0.97945
	Max.	1.
	Avg.	4.4005e-003

3.2.2 Strategies used for Mesh Iteration 2

Initial steps like part modelling, sweep method from Mesh Iteration 1 are retained for the rest of the iterations, primary focus while performing iteration 2 was to remove Trias from the mesh as only Quad/Hex elements are enough for heat conduction case. The following edge sizing and biases were given to the mesh:

- Global mesh size and copper face size is same as in iteration 1.
- Edge size of 0.003m on both edges of *Base_air* and suitable biasing was given to adjust the linearity of the mesh.
- Edge sizing of 0.0025m(wb-2, wb-3 & wb-4) and 0.002m(Inlet and Outlet BCs) was given to wall boundaries respectively.

3.2.3 Results for Mesh 2

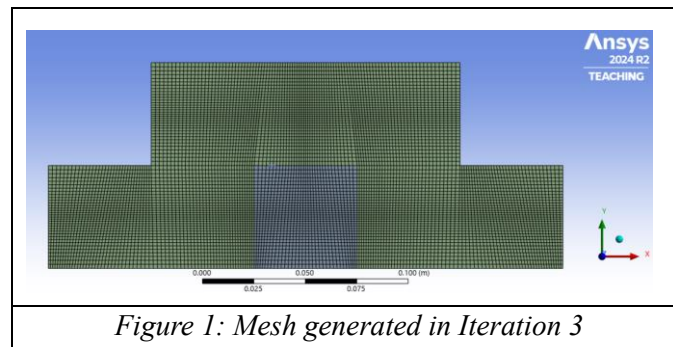


- Max. Temp: $5.7605e+02$
- Iterations: 6135
- Scaled residual: $1e-10$

Steps followed to ensure convergence:

- Brought the quality criteria within the limits to get accuracy in the results.
- Mesh elements set to all quad as the geometry is not much complex.
- Residual set to $1e-10$ to get a smoother graph and to check the convergence in the graph.

3.3 Mesh 3



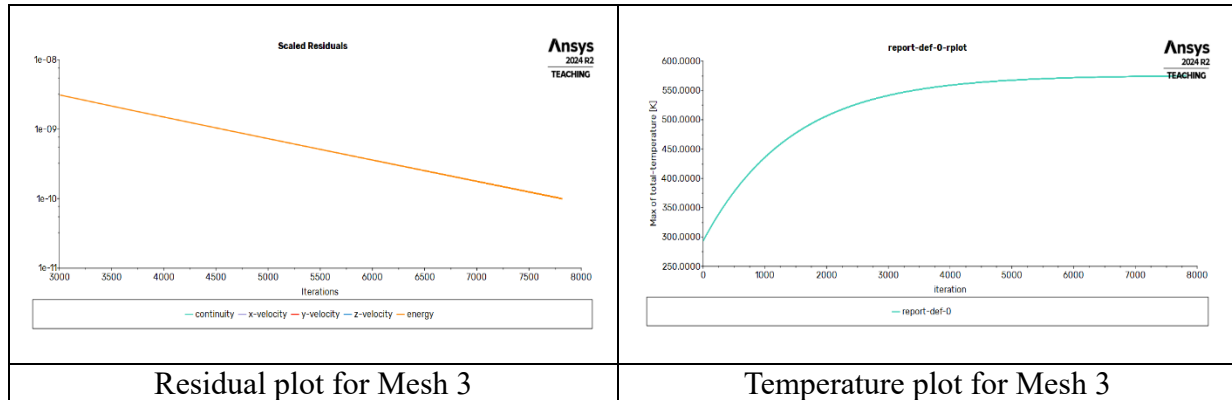
3.3.1 Details of Mesh 3

Mesh size (global)		0.003m
Mesh size (face sizing) copper block		0.002m
Elements		9112
Skew	Min.	$2.81e-004$
	Max.	$2.81e-004$
	Avg.	$2.81e-004$
Orthogonal Quality:	Min.	0.98616
	Max.	1.
	Avg.	0.99719

3.3.2 Strategies used for Mesh Iteration 3

- Edge sizing of 0.0015m is given on all wall boundaries and for both side of Base_air with biasing adjusted to smoothen the mesh.
- Edge sizing of 0.0015m given to Cu-2, Cu-3 & Cu-4 with suitable biasing factors.

3.3.3 Results for Mesh 3

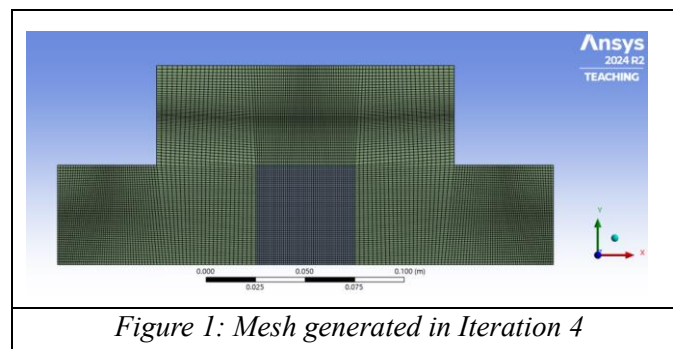


- Max. Temp: 5.7476e+02
- Iterations: 7817
- Scaled residual: 1e-10

Steps followed to ensure convergence:

- Reduced the element sizes and increased linearity in the mesh to get accuracy as well as faster results.
- Monitored maximum temperature plots to check consistency in the results.
- Residual set to 1e-10 to get a smoother graph and to check the convergence in the graph.

3.4 Mesh 4



3.4.1 Details of Mesh 4

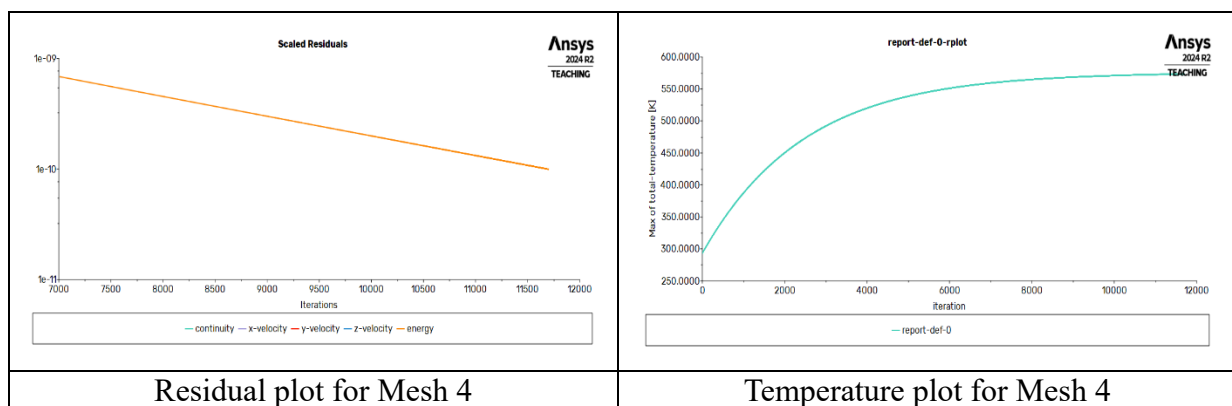
Mesh size (global)		0.002m
Mesh size (face sizing) copper block		0.001m
Elements		13824
Skew	Min.	1.3057e-010
	Max.	8.7454e-002
	Avg.	2.2539e-002

Orthogonal Quality:	Min.	0.99098
	Max.	1.
	Avg.	0.99879

3.4.2 Strategies used for Mesh Iteration 4

- Element and face sizes decreased from previous iterations to give a finer mesh and edge sizing adjusted to get a smooth linear mesh considering mesh connectivity with solid and fluid interface.
- Edge sizing of 0.0015m given to both edges of *Base_air* with suitable biasing factors.
- Edge sizing of 0.001 given to all wall boundaries and inlet and outlet BCs with adjusted biasing to decrease the skew factor and give a smoother mesh.

3.4.3 Results for Mesh 4

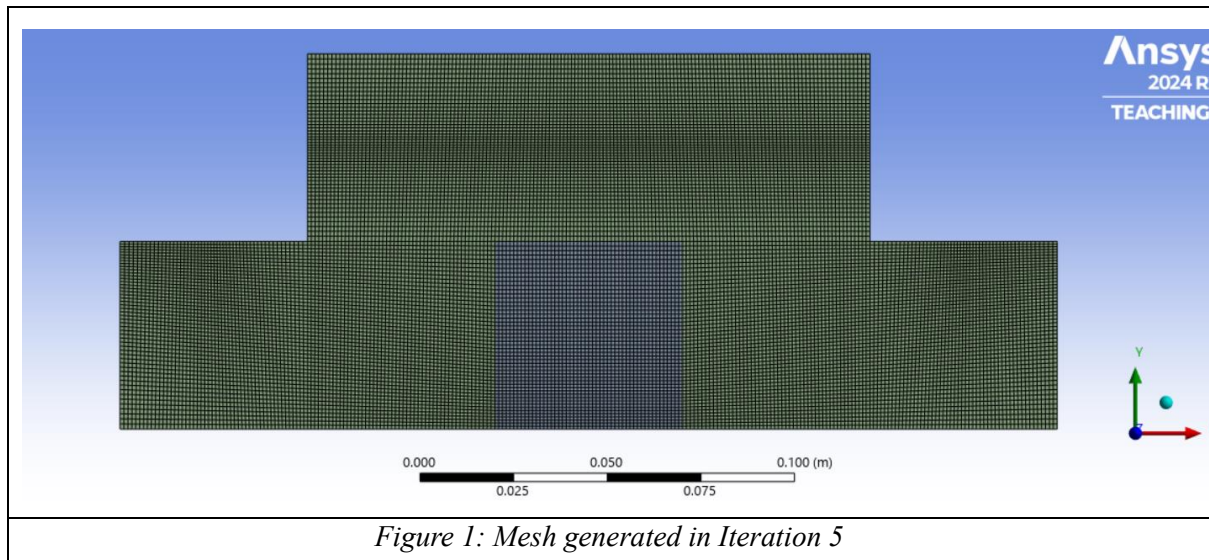


- Max. Temp: 5.73400E+02
- Iterations: 11699
- Scaled residual: 1e-10

Steps followed to ensure convergence:

- Refined the mesh by aligning the meshes more linearly and close to the copper block to get accuracy in the results.

3.5 Mesh 5



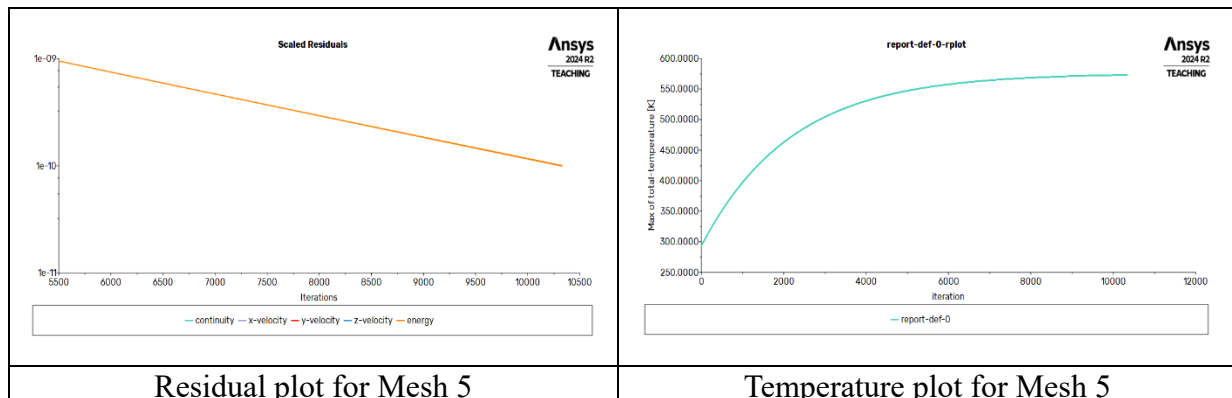
3.5.1 Details of Mesh 5

Mesh size (global)		0.0015m
Mesh size (face sizing) copper block		0.001m
Elements		20000
Skew	Min.	1.3057e-010
	Max.	6.6769e-002
	Avg.	2.0839e-002
Orthogonal Quality:	Min.	0.99451
	Max.	1.
	Avg.	1.1142e-003

3.5.2 Strategies used for Mesh Iteration 5

- Element and face sizes decreased further to check accuracy of the results while maintaining the skew angle within the considerable limits.
- Edge sizing of 0.001m given to both edges of *Base_air* with suitable biasing factors.
- Edge sizing of 0.001m given to all wall boundaries and inlet and outlet BCs with adjusted biasing to decrease the skew factor and give a smoother mesh.
- Edge sizing of 0.001m given on Cu_2 and Cu_4.

3.5.3 Results for Mesh 5



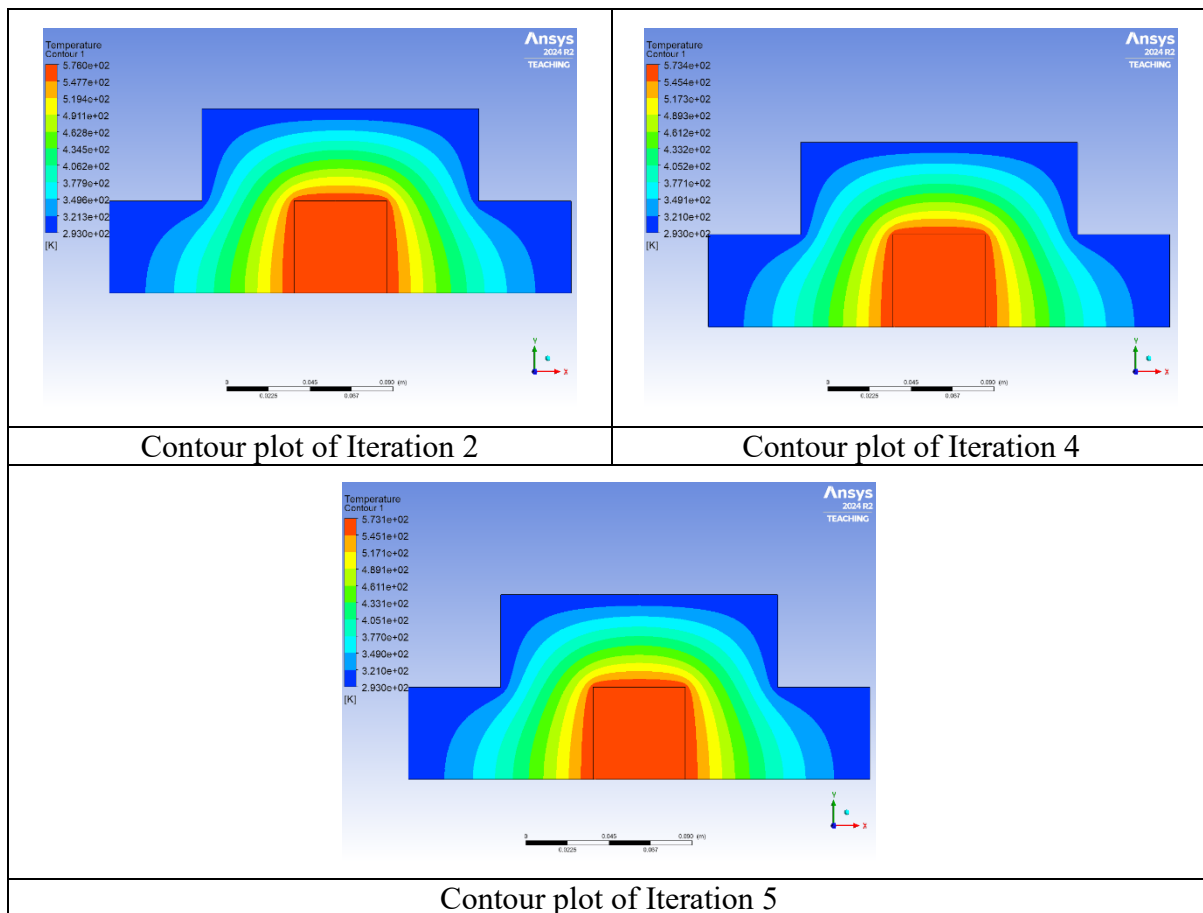
- Max. Temp: 5.73140E+02
- Iterations: 10329
- Scaled residual: 1e-10

Steps followed to ensure convergence:

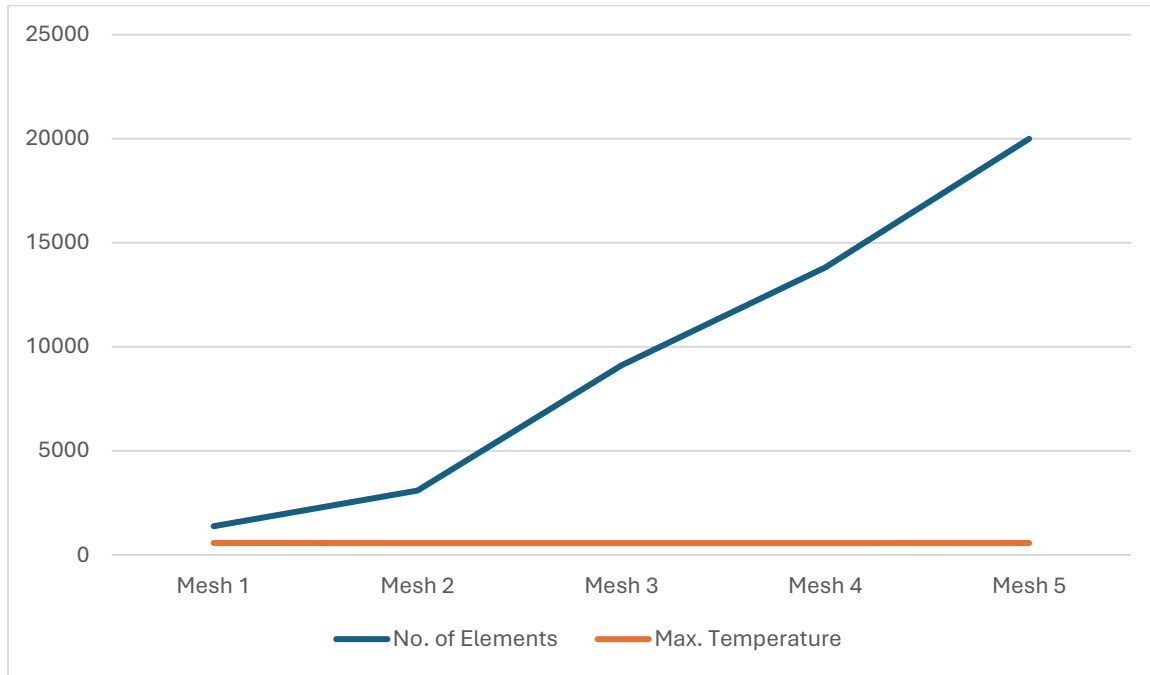
- Refined the mesh by aligning the meshes more linearly and close to the copper block to get accuracy in the results.
- Increased the number of elements to get total temperature accuracy, also decreasing the number of iterations to get the results.

4 Result Comparison

The contour plots of Mesh Iteration 2, 4 and 5 to compare the results between coarse to finer meshes.



Iteration	Global element size	No. of Elements	Max. Temperature	Scaled residuals
Mesh 1	0.005	1381	5.76390E+02	1.00E-10
Mesh 2	0.005	3099	5.76050E+02	1.00E-10
Mesh 3	0.003	9112	5.74760E+02	1.00E-10
Mesh 4	0.002	13824	5.73400E+02	1.00E-10
Mesh 5	0.0015	20000	5.73140E+02	1.00E-10



The graph shows that even with significant changes in the number of elements, the difference in consecutive results is minimum. Hence, it is possible to get a mesh independent result with fine refinement of mesh and keeping the elements within the 100,000 threshold.

5 Recommendations for Further Investigation

- Using the sweep method for structured meshes
- Refining meshes near high-gradient regions (Boundary Layers and Interfaces)
- Keeping the aspect ratio and element quality within the limits
- Checking convergence criteria for lesser residuals with finer meshes

6 Governing Equation for Heat Convection

Since heat convection is time dependent, the process varies over time and hence the governing equation for heat convection is:

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + Q$$

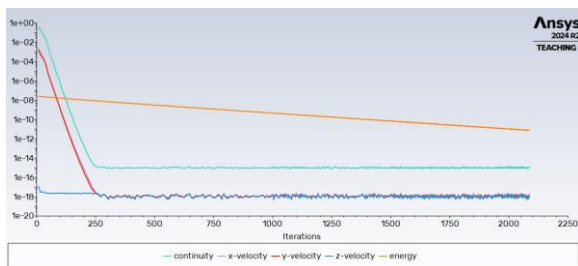
where:

- ρ = Density of the fluid (kg/m³)

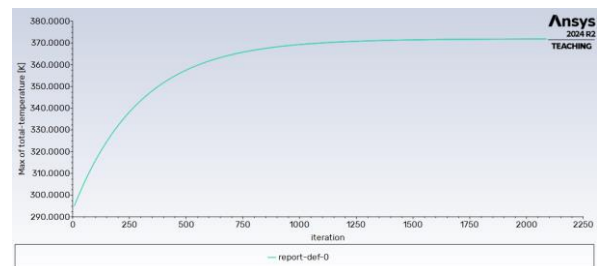
- ρC_p = Specific heat capacity (J/kg·K)
- T = Temperature (K)
- $\mathbf{v} \cdot \nabla T$ = Convective heat transport term (advection)
- k = Thermal conductivity (W/m·K)
- Q = Internal heat generation per unit volume (W/m³)
- $\nabla \cdot (k \nabla T)$ = Heat conduction term
- $\frac{\partial T}{\partial t}$ = Time-dependent term (for transient analysis)

7 Results of convection case

7.1 Mesh 1:

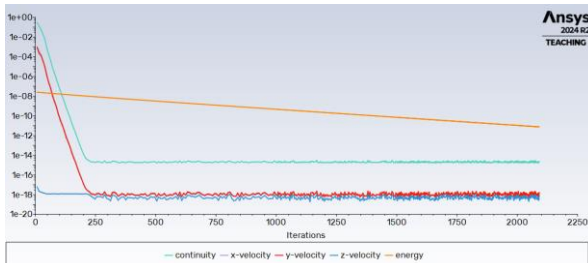


Scaled residual for mesh 1

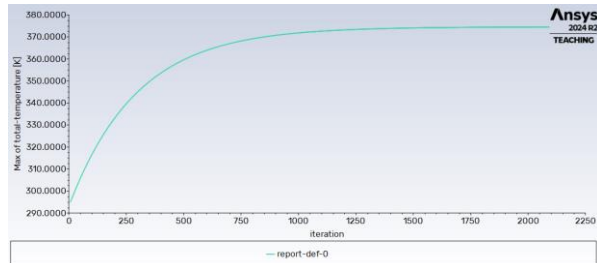


Temperature plot for mesh 1

7.2 Mesh 2:

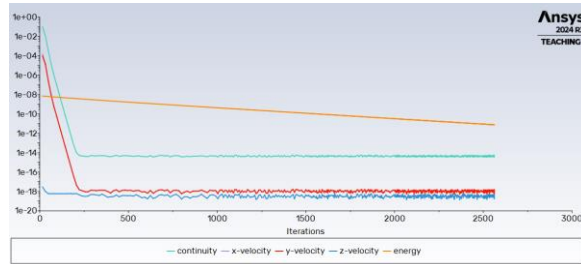


Scaled residual for mesh 2

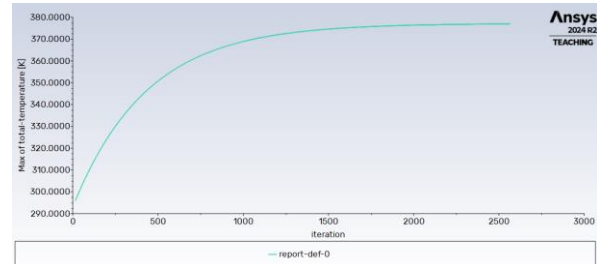


Temperature plot for mesh 2

7.3 Mesh 3:

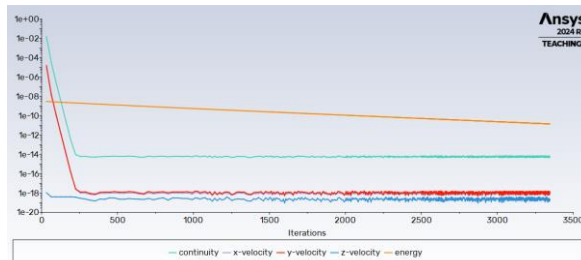


Scaled residual for mesh 3

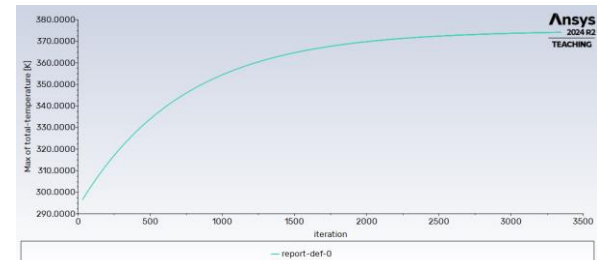


Temperature plot for mesh 3

7.4 Mesh 4:

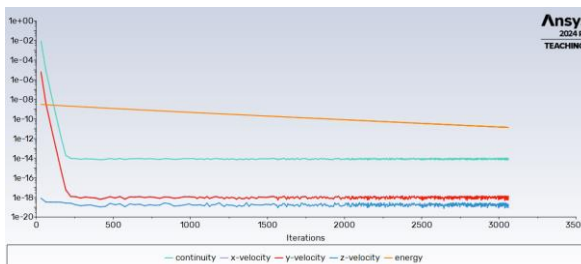


Scaled residual for mesh 4

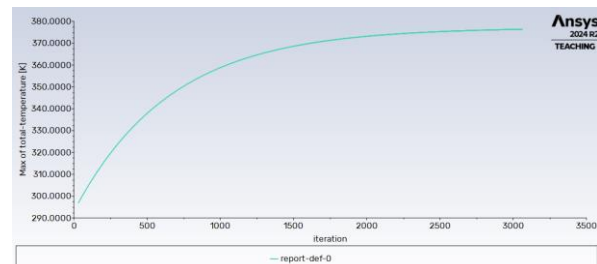


Temperature plot for mesh 4

7.5 Mesh 5:



Scaled residual for mesh 5



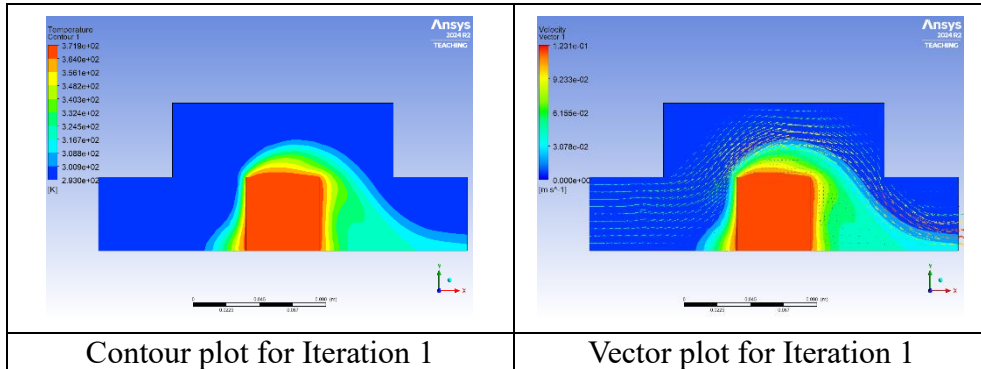
Temperature plot for mesh 5

- The maximum temperature increases rapidly in the first few iterations and gradually increases in the next set of meshes.
- After ~2500 iterations, the curve flattens, showing a stable solution.
- No significant fluctuations occur beyond this point, indicating temperature convergence.
- The results have converged successfully over shorter number of iterations.

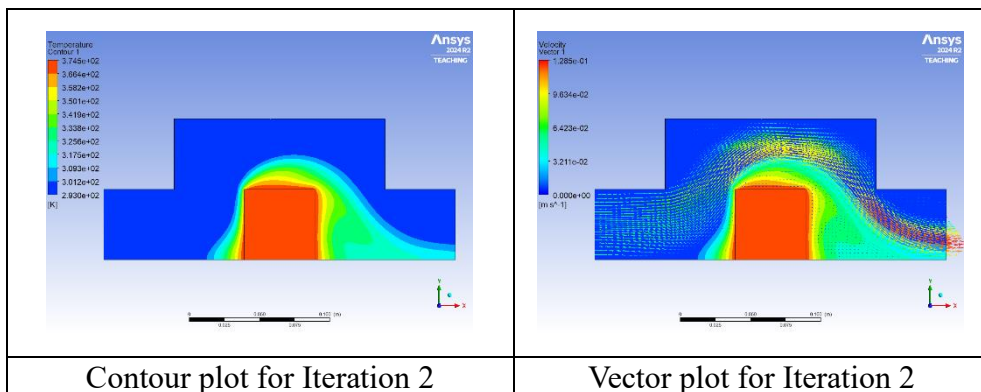
8 Contour and vector plots for convection

Velocity	Temperature(Convection)
1.23E-01	3.719E-02
1.29E-01	3.75E-02
1.29E-01	3.76E-02
1.31E-01	3.77E-02
1.30E-01	3.77E-02

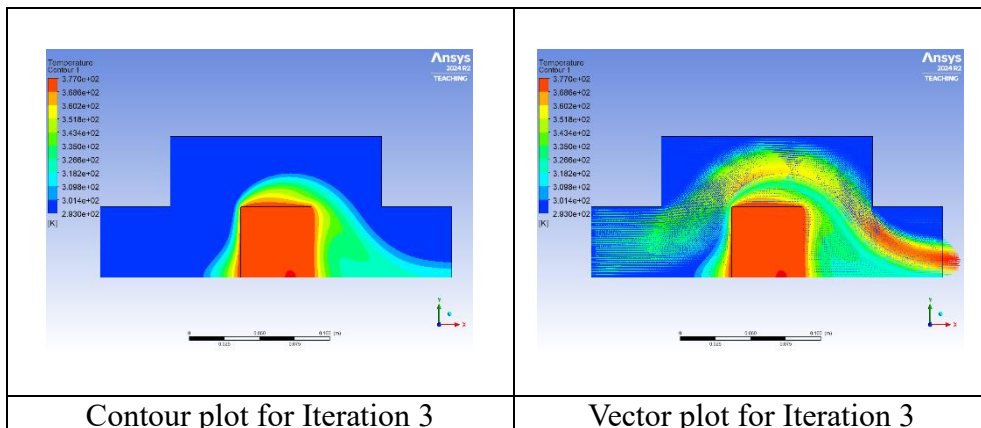
8.1 Iteration 1



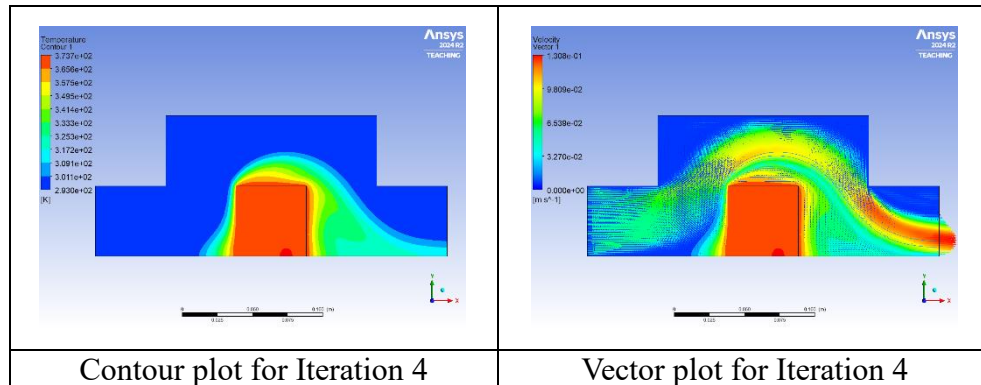
8.2 Iteration 2



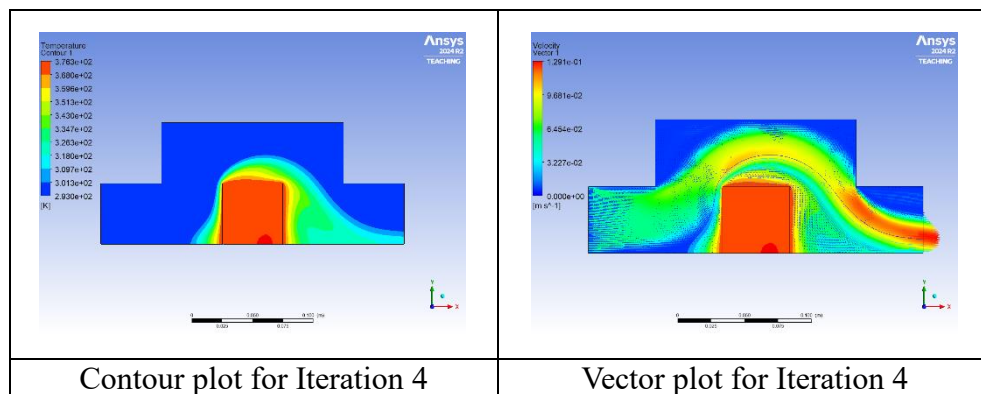
8.3 Iteration 3



8.4 Iteration 4



8.5 Iteration 5



- The temperature increase is gradual, suggesting the system is approaching a steady-state condition, where the heat transfer rate has stabilized.
- The small changes in temperature indicate that convection is occurring but not at an extreme rate, possibly due to low airflow velocity or a relatively low-temperature gradient.
- If the goal is to improve heat dissipation, a higher airflow velocity might be necessary to further enhance convective heat transfer.

9 Recommendations for further refinement

- Expand the velocity range: Testing higher airflow velocities (e.g., 0.1 m/s – 1.0 m/s) would show a more accurate effect on heat transfer.
- Analyse Reynolds Number: Checking whether the airflow is laminar or turbulent can help understand how effectively heat is being transferred.
- Surface Conditions: Investigating how different surface roughness or materials affect convective heat transfer.

A wall boundary acts like a solid surface that can transfer heat and blocks fluid flow, while a symmetry boundary treats the domain as a mirror, meaning no heat flows across it, no fluid moves through it, and it helps reduce the computational load if the system is symmetrical.

In convection case, a wall puts forward a no-slip condition, showing that the fluid velocity at the surface is null, which influences the boundary layer and the rate of heat transfer. The results are attached below:

